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DiffOSeg: Consensus and Preference

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1.1 Multi-rater segmentation



One CT patch · four expert annotations

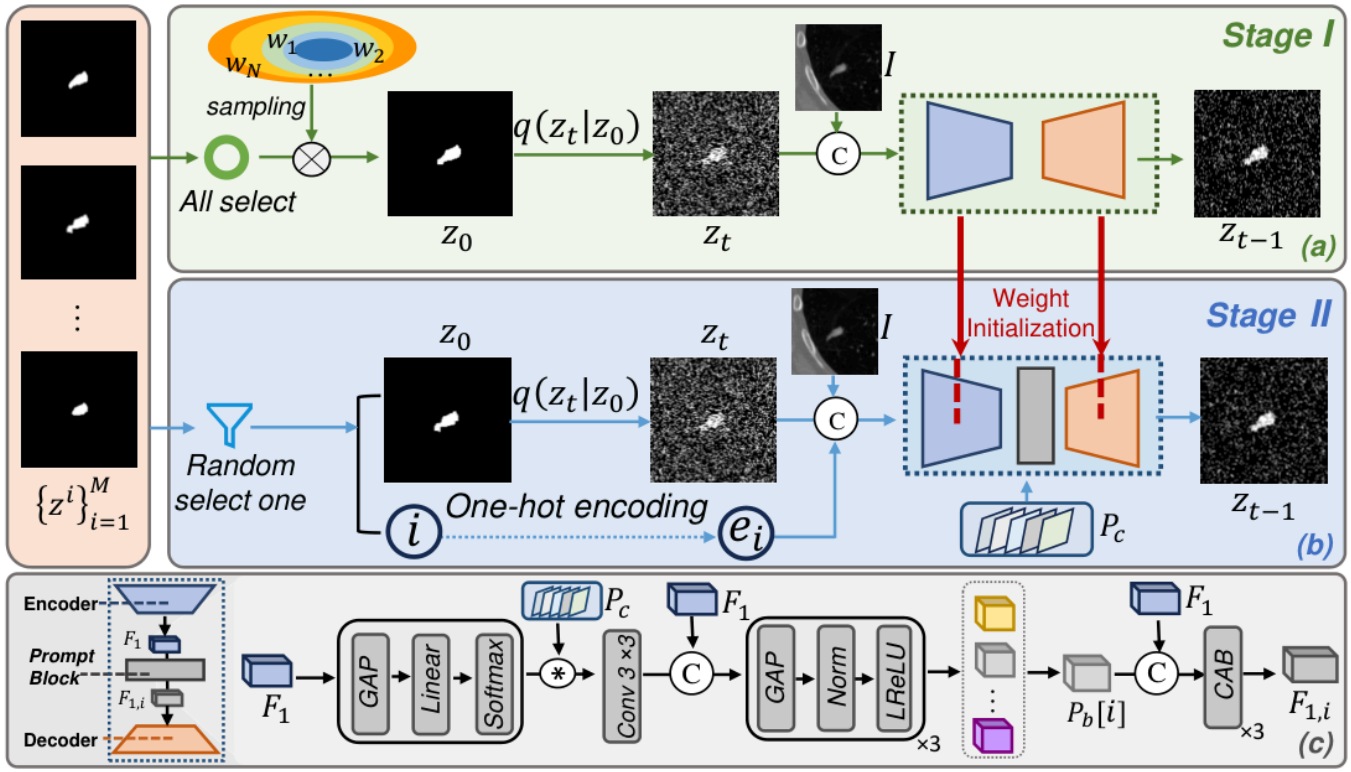
Consensus

Which segmentation patterns are shared, and where do annotations disagree?

Preference

How can a prediction reflect a particular annotation style?

1.2 DiffOSeg in two stages



Stage I

Image-conditioned categorical diffusion learns a distribution of masks.

Stage II

Initialize from Stage I; add adaptive preference prompts.

Our target: the Stage I consensus strategy.

1.3 P versus A

P: probabilistic consensus

Choose 1–4 annotations, then average the selected subset.

$$z_P = \frac{1}{|S|} \sum_{j \in S} z_j$$

The training target changes with the subset.

A: average aggregation

Average all four annotations on every visit.

$$z_A = \frac{1}{4} \sum_{j=1}^{4} z_j$$

The consensus probabilities stay fixed.

Frozen local choices: uniform subset size and subset; per-pixel categorical target draws in both arms. These operational details are disclosed assumptions.

Does P outperform A on LIDC-IDRI?

Table 3 · Stage I · scaled-faithful · one patient fold · three paired seeds

Endpoint	What it measures	Expected change
GED30	Mismatch between mask distributions	$P - A < 0$
DiceSoft30	Threshold-aware overlap of mean masks	$P - A > 0$

30 = generated masks per image. DiceSoft uses five thresholds.

Supported: both paired 95% intervals favor P.
Not supported: either metric is significantly reversed.
Inconclusive: all other outcomes.

2.2 Paired experiment

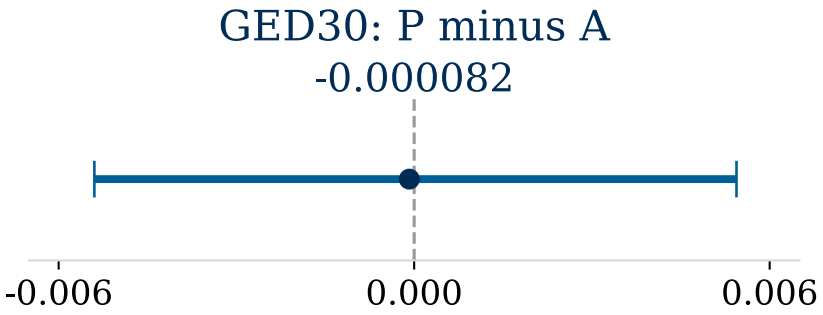
Design choice	Frozen local comparison
Patient separation	160 train / 54 test patients; 450 test patches
Repeated training	P and A × 3 paired seeds; matched initialization
Shared budget	120,000 updates; batch 12; final EMA
Preserved settings	128 × 128; 250 cosine steps; K = 30
Compute	12.157 GPU-hours used / 240-hour ceiling

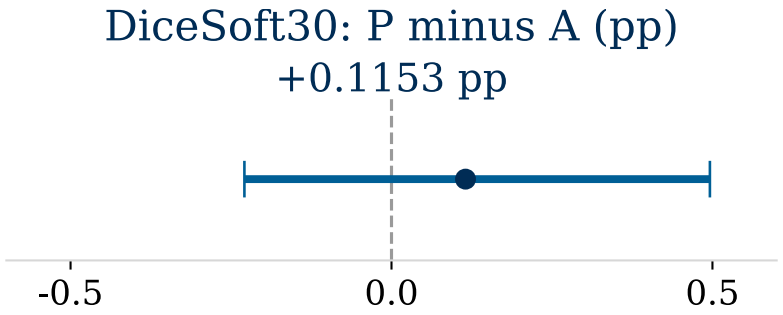
Paper: four folds. Local test: one fold, with frozen consensus encoding and augmentation assumptions.

Earlier P-only runs had already examined this cohort. The freeze predates this P/A comparison, not all prior exposure.

2.3 P/A result: inconclusive

Metric	Paper P / A	Local P / A
GED30 ↓	0.0773 / 0.1029	0.104276 / 0.104358
DiceSoft30 ↑	92.20 / 90.43%	90.3139 / 90.1986%





Both paired intervals cross zero: the declared outcome is inconclusive.

95% intervals use 10,000 paired seed-by-patient draws; patients retain all patches. Paper: four-fold mean. Local: three-seed mean on one fold.

2.4 Discrepancy diagnosis

Checks completed

- Matching split, budget and paired initialization
- Source and checkpoint identities checked
- Saved mask metrics and bootstrap reproduced

Explanations still open

- Consensus encoding and augmentation choices
- Split composition and training variability
- One-fold scale and prior cohort exposure

Under our sampler, $E[z_P] = z_A$ at each pixel; shared subsets can still change spatial dependence.

This mechanism does not identify the cause of the paper–local gap. An inconclusive result does not prove equivalence.

3.1 Improvement hypotheses

Reproduced DiffOSeg: four patient folds, one seed recipe, a different frozen training path. Historical primary endpoints: GED50, DiceSoft50 and preference Dice.

Hypothesis	Mechanism	Pilot → full evaluation
H1 · Boundary	Correct localization error	Inner gate passed → v1 improved
H2 · Polar	Model contour variation	Pilot qualified → v2 negative
H3 · Votes	Supervise pixel disagreement	Inner selection → v3 improved
H4 · Energy	Calibrate sample distribution	Decoder failed; readout passed → v4

We focus on v4, retaining its inherited components, failed alternatives and controls. This study does not replace the P/A test.

3.2 Energy calibration

Accurate marginal votes need not yield a good spatial distribution.

Step	Input ⟶ operation ⟶ output
1 · Joint decoding	Four rank heads + vote heads ⟶ 16 binary label states
2 · Distribution readout	Cumulative supports + proposal frequency ⟶ calibrated frequency
3 · Sampling	Calibrated frequency + sample ordering ⟶ K = 50 masks

Fit the readout with **empirical GED + 3 × DiceSoft loss**.

Calibration uses internal training-patient partitions; the readout is locked before outer inference. The decoder-only pilot failed its GED gate.

3.3 Results and controls

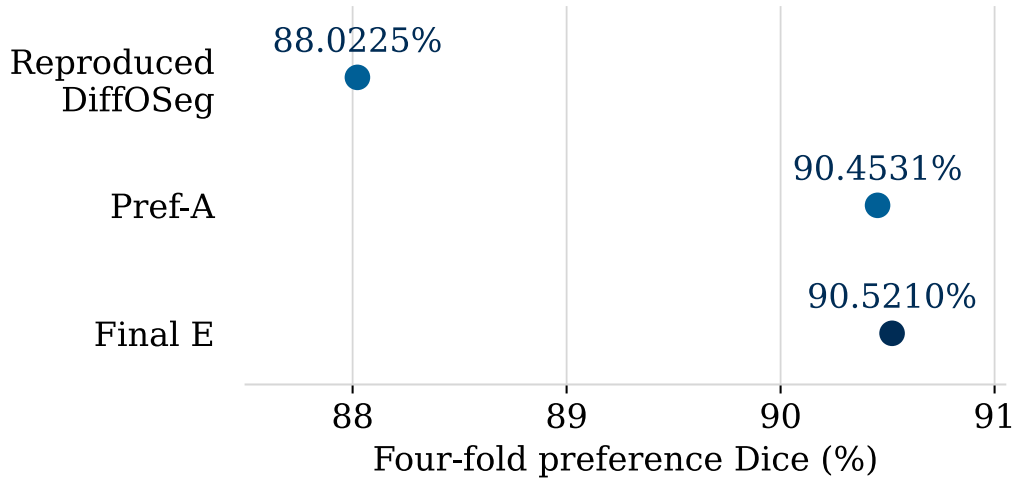
Variant	GED50 ↓	DiceSoft50 (%) ↑
Votes v3	0.107941	90.696
Full v4	0.104515	90.790
No energy readout	0.115012	90.730
Original sample order	0.103678	90.790
Energy-only on v3	0.097209	90.483

v4 – v3 GED50: –0.003426; 95% CI [–0.005314, –0.001551]

The old order improves GED further; energy-only trades DiceSoft for GED. The small v4 Dice gains have intervals crossing zero.

Controls reuse fitted weights. Patient intervals condition on those models; total compute is not matched.

3.4 Preference extension



90.5210%

+2.4985 pp vs Reproduced DiffOSeg

Localization / registration, followed by SAM adaptation and a three-seed SAM2.1 combination.

Extra models and compute; the historical 91% target remains unmet.

Four-fold preference Dice averages four area-ranked masks. Ranks are size-based styles, not persistent clinician identities.

Same repeatedly used cohort. Pref-A is an extension system, unrelated to A = average aggregation in the P/A test.

4.1 Agent use and evidence

Phase	Consequential action	Verification record
Understand / audit	Recovered the omitted P/A archive	Frozen source, six runs and pairing
Implement / check	Built CPU evidence checks	Metric fixtures and package checks
Analyze / challenge	Retained the inconclusive decision	Full saved-mask and bootstrap replay

Reported results are traceable; evidence checks passed.

4.2 Limits and next steps

Current limit	Next verification
P/A: one fold and only three seeds	More folds / seeds; isolate encoding assumptions
Repeated use of the LIDC cohort	Freeze the method and test a new cohort
Added models and incomplete compute totals	Account for costs; compare matched budgets
Extension CIs condition on fitted models	Repeat the complete training and selection process

A three-seed ensemble is not three independent system repetitions.

The P/A claim, four-fold consensus study and preference extension retain separate conclusions.

4.3 Conclusions

P/A reproduction: complete, but inconclusive

The local advantages are small and both paired intervals cross zero.

Energy calibration: a supported local GED gain

Negative results and component controls limit the explanation.

Preference extension: 90.5210% with added resources

Promising on this cohort; generalization and compute fairness remain open.